Learning Objective 3.6: I can distinguish between array factor and true antenna pattern and account for element pattern effects.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **How the kit builds its sweep.** The ADALM-PHASER carries $N = 8$ elements spaced $d = 14$ mm apart. At 10.3 GHz the wavelength is 29.1 mm, so $kd = 173.2^\circ$ of phase per unit of sin θ. In the Rectangular plot the kit holds the HB100 still and steps the element-to-element phase one 2.8125° LSB per point.

   (a) Find the change in $\sin\theta_0$ for one step, and the step in angle near broadside.

   $\Delta\theta_0 =$ ☐

   (b) Find the same step in angle near 30°.

   $\Delta\theta_0 =$ ☐

   (c) Every point of the sweep comes from a different beam. In two or three sentences, explain why the trace still looks like the pattern of a single beam.

   (d) State one thing the sweep cannot show that a measurement with one beam held still can, and give the reason.

2. **The walk.** With Lab Preset 7 the kit holds the beam at 0° and plots the peak level against time while you carry the HB100 from -90° to $+90^\circ$ on a 1.2 m arc in 25 s. The beam's half-power width is 13.3°.

   (a) At a constant pace, how long does the source spend inside the half-power width of the main lobe?

   t = ____

   (b) You walk through the main lobe at 70% of your average pace, then convert time to angle at the average pace. What half-power width does the trace report?

   θ_{HP} = ____

   (c) Does the slow pace in part (b) change the level of the main lobe or of the first sidelobes on the trace? Explain in one or two sentences.

   (d) Your arm drifts 10 cm outward, to 1.3 m, as you walk. By how much does the received level change?

   ΔL = ____

3. **Comparing the two traces.** A team records a sweep with the HB100 at boresight and a walk with the beam held at 0°, both at 1.2 m. The sweep reads its first sidelobes at -12.6 and -13.4 dBc. The walk reads them at -11.6 and -14.7 dBc. Each reading carries about ± 1 dB from room reflections.

   (a) Do the two methods agree on the first sidelobes? Give the reason, using the uncertainty of a dBc value.

   (b) The team reads the walk's main-lobe width as 17° and the sweep's as 13°. Which width do you report, and why?

   (c) Beyond $\pm 50^\circ$ the sweep's lobes read about -17 dBc and the walk's about -21 dBc. Name two reasons the far lobes differ.

   (d) The walk shows a lobe at -20 dBc near $+75^\circ$ that is not in the sweep, and it moves when a classmate steps aside. What is it?

4. **The beam at 30°.** The team now holds the beam at 30° for the walk, and places the HB100 at 30° for the sweep. For this array $\lambda/(Nd) = 0.260$ at 10.3 GHz.

   (a) Find the angles of the two first nulls of the 30° beam.

   θ_{null} = [ ]

   (b) Explain why the main lobe reaches farther above its peak than below it, and whether the sweep and the walk should both show this.

   (c) The array factor puts both first sidelobes of this beam at -12.8 dBc. The walk reads the outer one, near 60°, at about -15 dBc, while the sweep reads it near -12 dBc. Give the cause, and say whether one walk in a classroom can confirm it.

5. **Source distance.** Lesson 18 takes the course array's aperture as $D = Nd = 112$ mm. You place the HB100 1.2 m from the array face.

   (a) Compute the far-field distance for this array at 10.3 GHz ($\lambda = 29.1$ mm), and state whether the lab range qualifies.

   r =

   (b) You need to measure a larger array on the same 1.2 m range at 10.3 GHz. Find the largest aperture dimension that still satisfies the far-field criterion.

   D =

   (c) Name two changes to the setup, not the antenna, that make the far sidelobes of either trace more trustworthy.

Documentation: _____